Planar Graphs in Blowups of Fans*

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Abstract

We show that every n-vertex planar graph is contained in the graph obtained from a fan by blowing up each vertex by a complete graph of order $O(\sqrt{n}\log^2 n)$. Equivalently, every n-vertex planar graph G has a set X of $O(\sqrt{n}\log^2 n)$ vertices such that $G - X$ has bandwidth $O(\sqrt{n}\log^2 n)$. This result holds in the more general setting of graphs contained in the strong product of a bounded treewidth graph and a path, which includes bounded genus graphs, graphs excluding a fixed apex graph as a minor, and k-planar graphs for fixed k. These results are obtained using two ingredients. The first is a new local sparsification lemma, which shows that every n-vertex planar graph G has a set of $O((n\log n)/D)$ vertices whose removal results in a graph with local density at most D. The second is a generalization of a method of Feige and Rao, that relates bandwidth and local density using volume-preserving Euclidean embeddings.

1 Introduction

This paper studies the global structure of planar graphs and more general graph classes, through the lens of graph blowups. Here, the *b-blowup* of a graph H is the graph obtained by replacing each vertex v of H with a complete graph K_v of order b and replacing each edge vw of H with a complete bipartite graph with parts $V(K_v)$ and $V(K_w)$, as illustrated in Figure 1. We consider the following question: What is the simplest family of graphs $\mathcal{H}$ such that, for each n-vertex planar graph G, there is a graph $H \in \mathcal{H}$ such that G is contained[1] in a $\tilde{O}(\sqrt{n})$-blowup of H, where $\tilde{O}$ notation hides polylog(n) terms? We show that one can take $\mathcal{H}$ to be the class of fans, where a *fan* is a graph consisting of a path P plus one *center* vertex adjacent to every vertex in P.

THEOREM 1. *For each $n \in \mathbb{N}_1$ there exists a $O(\sqrt{n}\log^2 n)$-blowup of a fan that contains every n-vertex planar graph.*[2]

The blowup factor in this result is close to best possible. The k-blowup of a graph H with treewidth t has treewidth at most $k(t+1)-1$.[3] Since there are n-vertex planar graphs of treewidth $\Omega(\sqrt{n})$ (such as the $\sqrt{n}\times\sqrt{n}$ grid), any result like Theorem 1 that finds all planar graphs in blowups of bounded treewidth graphs must have

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[1]We say that a graph G is *contained* in a graph G' if G is isomorphic to a subgraph of G'.

[2]$\mathbb{N}$ denotes the set of non-negative integers and $\mathbb{N}_1 := \mathbb{N}\setminus\{0\}$ denotes the set of positive integers.

[3]A *tree-decomposition* of a graph G is a collection $(B_x)_{x\in V(T)}$ of subsets of $V(G)$ indexed by a tree T, such that: (a) for every edge $vw \in E(G)$, there exists a node $x \in V(T)$ with $v, w \in B_x$, and (b) for every vertex $v \in V(G)$, the set $\{x \in V(T) : v \in B_x\}$ induces a non-empty (connected) subtree of T. The *width* of such a tree-decomposition is $\max\{|B_x| : x \in V(T)\} - 1$. A *path-decomposition* is a tree-decomposition where the underlying tree is a path, denoted by the corresponding sequence of bags. The *treewidth* $\mathrm{tw}(G)$ of a graph G is the minimum width of a tree-decomposition of G. The *pathwidth* $\mathrm{pw}(G)$ of a graph G is the minimum width of a path-decomposition of G. Treewidth is the standard measure of how similar a graph is to a tree. Pathwidth is the standard measure of how similar a graph is to a path. By definition, $\mathrm{tw}(G) \leqslant \mathrm{pw}(G)$ for every graph G.

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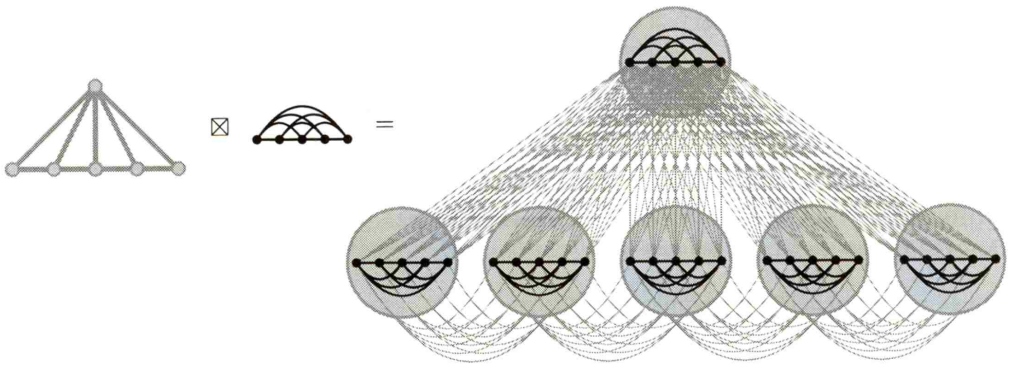


Figure 1: The 5-blowup of a 6-vertex fan.

blowups of size $\Omega(\sqrt{n})$ (and fans have treewidth 2, in fact pathwidth 2). Several other aspects of Theorem 1 are best-possible. These are discussed in Section 1.2, after a review of related previous work.

Theorem 1 can be restated in terms of the following classical graph parameter. Let G be a graph. For an ordering $v_1, \dots, v_n$ of $V(G)$, let the *bandwidth* $\mathrm{bw}_G(v_1, \dots, v_n) := \max(\{|j-i| : v_iv_j \in E(G)\} \cup \{0\})$. The *bandwidth* of G is $\mathrm{bw}(G) := \min\{\mathrm{bw}_G(v_1, \dots, v_n) : v_1, \dots, v_n \text{ is an ordering of } V(G)\}$. See [1, 7, 8, 11, 25, 33, 36] for a handful of important references on this topic. It is well known that bandwidth is closely related to blowups of paths [7, 21]. Indeed, Theorems 1 and 2 are equivalent, where X is the set of vertices mapped to the center of the fan; see Lemmas 1.1 and 1.2 for a proof.

THEOREM 2. *Every n-vertex planar graph G has a set X of $O(\sqrt{n}\log^2 n)$ vertices such that $G - X$ has bandwidth $O(\sqrt{n}\log^2 n)$.*

We in fact prove several generalizations of Theorem 2 that (a) study the tradeoff between $|X|$ and the bandwidth of $G - X$, and (b) consider more general graph classes than planar graphs.

It follows from the Lipton–Tarjan Planar Separator Theorem that every n-vertex planar graph G satisfies $\mathrm{tw}(G) \leqslant \mathrm{pw}(G) \in O(\sqrt{n})$ (see [5]). It is also well known that $\mathrm{pw}(G) \leqslant \mathrm{bw}(G)$ (since if $\mathrm{bw}_G(v_1, \dots, v_n) \leqslant k$ then $\{v_1, \dots, v_{k+1}\}, \{v_2, \dots, v_{k+2}\}, \dots, \{v_{n-k}, \dots, v_n\}$ is a path-decomposition of G of width k). However, $\mathrm{bw}(G) \in \tilde{O}(\sqrt{n})$ is a much stronger property than $\mathrm{pw}(G) \in \tilde{O}(\sqrt{n})$. Indeed, bandwidth can be $\Omega(n/\log n)$ for very simple graphs, such as n-vertex complete binary trees. This highlights the strength of Theorem 2. In fact, Theorem 2 is tight (up to polylog factors) even for complete binary trees. For a complete binary tree T on n vertices, $\mathrm{bw}(T) \in \Omega(n/\log n)$ since the root of T is within distance $\log n$ of all vertices. For any set $X \subseteq V(T)$, T contains a complete binary tree with $\Omega(n/|X|)$ vertices that avoids X, so $\mathrm{bw}(T - X) \geqslant \Omega((n/|X|)/\log(n/|X|))$. Thus, $\mathrm{bw}(T - X) \in \tilde{\Omega}(\sqrt{n})$ for any set $X \subseteq V(T)$ of size $\tilde{O}(\sqrt{n})$.

1.1 Previous Results As summarized in Table 1, we now compare Theorem 1 with results from the literature, starting with the celebrated *Planar Separator Theorem* due to Lipton and Tarjan [31], which states that any n-vertex planar graph G contains a set X of $O(\sqrt{n})$ vertices such that each component of $G - X$ has at most $n/2$ vertices. This theorem quickly leads to results about the blowup structure of planar graphs. By applying it recursively, it shows that any n-vertex planar graph G is contained in a graph that can be obtained from the closure of a tree of height $O(\log n)$ by blowing up the nodes of depth i into cliques of size $O(\sqrt{n/2^i})$. (This observation is made by Babai, Chung, Erdős, Graham, and Spencer [3] to show that there is a *universal graph* with $O(n^{3/2})$ edges and that contains all n-vertex planar graphs.) By applying it differently, Lipton and Tarjan [32] show that G is contained in a graph obtained from a star by blowing up the root into a clique of size n^{1-a} and blowing up each leaf into a clique of size $O(n^{2a})$. These two structural results have had an enormous number of applications for algorithms, data structures, and combinatorial results on planar graphs. The second result, with $a = n^{1/3}$, shows that G is contained in a $O(n^{2/3})$-blowup of a star. Dvořák and Wood [23] use the second result recursively (with the size of the separator fixed at $c\sqrt{n}$ even for subproblems of size less than n) to show

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that G is contained in the $O(\sqrt{n})$-blowup of the closure of a tree of height $O(\log \log n)$. That is, G is contained in the $O(\sqrt{n})$-blowup of a graph of treedepth $O(\log \log n)$. The same method, with the size of the separator fixed at $cn^{1/2+\epsilon}$ shows that G is contained in an $O(n^{1/2+\epsilon})$-blowup of a graph of treedepth $O(1/\epsilon)$ [23].

Table 1: Results on b-blowups of a graph H that contain every n-vertex graph G from graph class $\mathcal{G}$.

class $\mathcal{G}$	H	lower bound on b		upper bound on b	
planar	tree	$\Omega(n^{2/3})$	[30]	$O(n^{2/3})$	[32]
	$\mathrm{tw} \leqslant 2$	$\Omega(\sqrt{n})$		$O(\sqrt{n})$	[15]
	fan	$\Omega(\sqrt{n})$		$O(\sqrt{n}\log^2 n)$	Theorem 1
	$\mathrm{tw} \leqslant 3$	$\Omega(\mathrm{tw}(G))$		$\mathrm{tw}(G)+1$	[27]
	$\mathrm{td} \leqslant c \log \log n$	$\Omega(\sqrt{n})$	[23]	$O(\sqrt{n})$	[23]
	$\mathrm{td} \leqslant c/\epsilon$	$\Omega(n^{1/2+\epsilon})$	[23]	$O(n^{1/2+\epsilon})$	[23]
max-degree Δ	tree	$\Omega(\Delta \cdot \mathrm{tw}(G))$	[38]	$O(\Delta \cdot \mathrm{tw}(G))$	[13, 18, 38]
K_t-minor-free	$\mathrm{tw} \leqslant t-2$	$\Omega(\sqrt{n})$		$O(\sqrt{tn})$	[27]
K_t-minor-free	$\mathrm{tw} \leqslant t-2$			$\mathrm{tw}(G)+1$	[27]
K_t-minor-free	$\mathrm{tw} \leqslant 4$	$\Omega(t\sqrt{n})$	[2]	$O_t(\sqrt{n})$	[15]
Euler genus g	fan	$\Omega(\sqrt{gn})$	[26]	$O(\sqrt{gn}+\sqrt{n}\log^2 n)$	Theorem 3
	$\mathrm{tw} \leqslant 2$	$\Omega(\sqrt{gn})$	[26]	$O((g+1)\sqrt{n})$	[15]
	$\mathrm{tw} \leqslant 3$			$2(g+1)(\mathrm{tw}(G)+1)$	[27]
(g,k)-planar	$\mathrm{tw} \leqslant O(k^3)$			$O(k^{3/4}g^{3/4}\sqrt{n})$	[23]
(g,k)-planar	$\mathrm{tw} \leqslant 10^8$	$\Omega(\sqrt{gkn})$	[19]	$O(c_{g,k}\sqrt{n})$	[17, 23]
k-planar	fan	$\Omega(\sqrt{kn})$	[19]	$O(k^{5/4}\sqrt{n}\log^2 n)$	Theorem 4
(g,k)-planar	fan	$\Omega(\sqrt{gkn})$	[19]	$O(g^{1/2}k^{5/4}\sqrt{n}\log^2 n)$	Theorem 5
$K_{3,t}$-minor-free	$\mathrm{tw} \leqslant 2$	$\Omega(\sqrt{tn})$		$O(t\sqrt{n})$	[15]
row treewidth t	fan	$\Omega(\sqrt{tn})$		$O(\sqrt{tn}\log^2 n)$	Theorem 6

Using different methods, Illingworth et al. [27] show that every n-vertex planar graph is contained in a $O(\sqrt{n})$-blowup of a graph with treewidth 3. Improving this result, Distel et al. [15] show that every n-vertex planar graph is contained in a $O(\sqrt{n})$-blowup of a treewidth-2 graph. They ask whether every planar graph is contained in a $O(\sqrt{n})$-blowup of a bounded pathwidth graph. Since fans have pathwidth 2, Theorem 1 answers this question, with $O(\sqrt{n})$ replaced by $O(\sqrt{n}\log^2 n)$.

Except for the star result (which requires an $\Omega(n^{2/3})$ blowup), all of the above results require blowing up a graph with many high-degree vertices. Theorem 1 shows that a pathwidth-2 graph with one high-degree vertex is enough, and with a quasi-optimal blowup of $O(\sqrt{n}\log^2 n)$. Thus, Theorem 1 offers a significantly simpler structural description of planar graphs than previous results.

A related direction of research, introduced by Campbell, Clinch, Distel, Gollin, Hendrey, Hickingbotham, Huynh, Illingworth, Tamitegama, Tan, and Wood [9], involves showing that every planar graph G is contained in the b-blowup of a bounded treewidth graph, where b is a function of the treewidth of G. They define the *underlying treewidth* of a graph class $\mathcal{G}$ to be the minimum integer k such that for some function f every graph $G \in \mathcal{G}$ is contained in a $f(\mathrm{tw}(G))$-blowup of a graph H with $\mathrm{tw}(H) \leqslant k$. They show that the underlying treewidth of the

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class of planar graphs equals 3. In particular, every planar graph G with $\mathrm{tw}(G) \leqslant t$ is contained in a $O(t^2 \log t)$-blowup of a graph with treewidth 3. Illingworth et al. [27] reduce the blowup factor to $t+1$. In this setting, treewidth 3 is best possible: Campbell et al. [9] show that for any function f, there are planar graphs G such that if G is contained in a $f(\mathrm{tw}(G))$-blowup of a graph H, then H has treewidth at least 3.

Allowing blowups of size $O(\sqrt{n})$ enables substantially simpler graphs H, since Distel et al. [15] show that $\mathrm{tw}(H) \leqslant 2$ suffices in this $O(\sqrt{n})$-blowup setting. Allowing for an extra $O(\log^2 n)$ factor in the blowup, the current paper goes further and shows that a fan H (which has pathwidth 2) suffices. For K_t-minor-free graphs (which also have treewidth $O_t(\sqrt{n})$ [2]), there is a similar distinction between $f(\mathrm{tw}(G))$-blowups and $O_t(\sqrt{n})$-blowups. Campbell et al. [9] show that the underlying treewidth of the class of K_t-minor-free graphs equals $t-2$, whereas Distel et al. [15] show that $\mathrm{tw}(H) \leqslant 4$ suffices for $O_t(\sqrt{n})$-blowups of H.

1.2 Optimality We now explain why, except possibly for the $\log^2 n$ factor, Theorem 1 cannot be strengthened. As already discussed above, a factor of $\sqrt{n}$ in the size of the blowup is necessary, since there are n-vertex planar graphs of treewidth $\Omega(\sqrt{n})$.

Pathwidth 2 is also the best possible bound in results like Theorem 1. Indeed, even *treewidth* 1 is not achievable: Linial et al. [30] describe an infinite family of n-vertex planar graphs G such that every (improper) 2-colouring has a monochromatic component on $\Omega(n^{2/3})$ vertices. Say G is contained in a b-blowup $(K_v : v \in V(T))$ of a tree T. Colour each vertex in each K_v by the colour of v in a proper 2-colouring of T. So each monochromatic component is contained in some K_v, implying that $b \in \Omega(n^{2/3})$.

Any graph of treedepth c has pathwidth at most $c-1$, so it is natural to ask if Theorem 1 can be strengthened to show that every n-vertex planar graph is contained in a $\tilde{O}(\sqrt{n})$-blowup of a bounded treedepth graph. The answer is no, as we now explain. Dvořák and Wood [23, Theorem 19] show that, for any $c \geqslant 1$ there exists $\epsilon > 0$ such that if the $\sqrt{n} \times \sqrt{n}$ grid is contained in a b-blowup of a graph H with treedepth at most c, then $b \in \Omega(n^{1/2+\epsilon})$. Thus, the $\sqrt{n} \times \sqrt{n}$-grid is not contained in a $\tilde{O}(n^{1/2})$-blowup of a graph with bounded treedepth. In particular, Theorem 1 cannot be strengthened to the treedepth setting without increasing the size of the blowup by a polynomial factor.

1.3 Graphs on Surfaces and With Crossings Theorem 1 generalizes to graphs embeddable on arbitrary surfaces as follows. Here, the *Euler genus* of a surface obtained from a sphere by adding h handles and c crosscaps is $2h+c$. The *Euler genus* of a graph G is the minimum Euler genus of a surface in which G embeds without crossings.

THEOREM 3. *For each $g, n \in \mathbb{N}_1$ there exists a $O(\sqrt{gn} + \sqrt{n}\log^2 n)$-blowup of a fan that contains every n-vertex graph of Euler genus at most g.*

Theorem 1 also generalizes to graphs that can be drawn with a bounded number of crossings on each edge. A graph G is *k-planar* if it has a drawing in the plane in which each edge participates in at most k crossings, and no three edges cross at the same point. This topic is important in the graph drawing literature; see [29] for a survey just on the $k=1$ case. We prove the following generalization of Theorem 1:

THEOREM 4. *For each $k, n \in \mathbb{N}_1$ there exists a $O(k^{5/4}\sqrt{n}\log^2 n)$-blowup of a fan that contains every n-vertex k-planar graph.*

We in fact prove the following generalization of Theorems 1, 3 and 4. Here a graph G is *(g,k)-planar* if it has a drawing in a surface of Euler genus at most g in which each edge is in at most k crossings, and no three edges cross at the same point.

THEOREM 5. *For each $g, k, n \in \mathbb{N}_1$ there exists a $O(k^{3/4}g^{1/2}n^{1/2} + k^{5/4}n^{1/2}\log^2 n)$-blowup of a fan that contains every n-vertex (g,k)-planar graph.*

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1.4 Subgraphs of $H \boxtimes P$ Theorems 1 and 3–5 each follow from a more general theorem about subgraphs of certain strong graph products, as we now explain. The *strong product* $A \boxtimes B$ of two graphs A and B is the graph with vertex set $V(A \boxtimes B) := V(A) \times V(B)$ that contains an edge with endpoints (v_1, v_2) and (w_1, w_2) if and only if

1. $v_1 w_1 \in E(A)$ and $v_2 = w_2$;
2. $v_1 = w_1$ and $v_2 w_2 \in E(B)$; or
3. $v_1 w_1 \in E(A)$ and $v_2 w_2 \in E(B)$.

Note that the k-blowup of H can be written as the strong product $H \boxtimes K_k$. Therefore, Theorem 1 states that for every n-vertex planar graph G there is a fan F such that G is isomorphic to a subgraph of $F \boxtimes K_{O(\sqrt{n}\log^2 n)}$.

The *row treewidth* of a graph G is the minimum integer t such that G is contained in $H \boxtimes P$ for some graph H with treewidth t and for some path P. We prove the following more general result:

THEOREM 6. *For every $t, n \in \mathbb{N}_1$, there exists a $O(\sqrt{tn}\log^2 n)$-blowup of a fan that contains every n-vertex graph with row treewidth at most t.*

Although we give more direct proofs of Theorems 1 and 3–5, Theorem 6 implies variants of all of these (with larger dependencies on g and k) because each of these classes of graphs has bounded row treewidth [16, 17, 20, 22, 37].[4] Very recently, Distel [14] has used Theorem 6 to prove a version of Theorem 1 that holds for any proper minor-closed family of graphs.

1.5 Bandwidth and Fan-Blowups The following straightforward lemma shows that Theorem 2 implies Theorem 1, and that each of Theorems 3–6 are implied by the analogous statements about bandwidth.

LEMMA 1.1. *For every $b, n \in \mathbb{N}$ with $1 \leqslant b \leqslant n$, let $\mathcal{G}$ be the class of n-vertex graphs such that $\mathrm{bw}(G - X) \leqslant b$ for some $X \subseteq V(G)$ with $|X| \leqslant b$. Let F be the fan on $\lceil n/b \rceil$ vertices. Then the b-blowup of F contains every graph in $\mathcal{G}$.*

Proof. Let $p := \lceil n/b \rceil - 1$. Let F be a fan with center r, where $F - r$ is the path $u_1, \dots, u_p$. So $|V(F)| = p + 1 = \lceil n/b \rceil$. Let U be the b-blowup of F.

Let $G \in \mathcal{G}$. So $|V(G)| = n$ and $\mathrm{bw}(G - X) \leqslant b$ for some $X \subseteq V(G)$ with $|X| \leqslant b$. Move vertices from $G - X$ into X so that $|X| = b$. Still $\mathrm{bw}(G - X) \leqslant b$. Let $v_1, \dots, v_{n-b}$ be an ordering of $G - X$ with bandwidth at most b. Injectively map X to the blowup of r. For $i \in \{1, \dots, p-1\}$, injectively map $v_{(i-1)b+1}, \dots, v_{ib}$ to the blowup of u_i. And injectively map $v_{(p-1)b+1}, \dots, v_{n-b}$ to the blowup of u_p. By construction, G is contained in U. □

REMARK 1. *The number of vertices in the fan-blowup U in Lemma 1.1 is $b\lceil n/b \rceil$. When mapping an n-vertex graph G to U, the $b - (n \bmod b)$ vertices of U not used in the mapping come from the blowup of u_p. By removing these vertices, we obtain a subgraph U_n of U with exactly n vertices that contains every graph in $\mathcal{G}$. One consequence of this is the following strengthening of Theorem 1: For each $n \in \mathbb{N}_1$, there exists an n-vertex subgraph U_n of a $O(\sqrt{n}\log^2 n)$ blowup of a fan that contains every n-vertex planar graph. Each of Theorems 3–6 has a similar strengthening.*

The next lemma provides a converse to Lemma 1.1.

LEMMA 1.2. *If an n-vertex graph G is contained in a b-blowup of a fan F, then $\mathrm{bw}(G - X) \leqslant 2b - 1$ for some $X \subseteq V(G)$ with $|X| \leqslant b$.*

Proof. Let r be the center of F. Let X be the set of vertices of G mapped to the blowup of r. So $|X| \leqslant b$. By definition, $P := F - r$ is a path. Let B_i be the set of vertices mapped to the blowup of the i-th vertex of P. Any ordering of $V(G)$ that places all vertices of B_i before those in B_{i+1} for each i has bandwidth at most $2b - 1$. Thus $\mathrm{bw}(G - X) \leqslant 2b - 1$. □

[4] In the case of k-planar, genus-g, and (g, k)-planar graphs the row treewidth is bounded by a function of g and/or k.

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1.6 Techniques Throughout the paper, we use $\log(x)$ for the base-2 logarithm of x, and we use $\ln(x)$ for the natural logarithm of x. When a logarithm appears inside O-notation, we use the convention $\log(x) := 1$ for all $x \leqslant 2$.

For a graph G and any two vertices $v, w \in V(G)$, define the *(graph) distance* between v and w, denoted $d_G(v, w)$, as the minimum number of edges in any path in G with endpoints v and w or define $d_G(v, w) := \infty$ if v and w are in different components of G. For any $r \geqslant 0$ and any $v \in V(G)$, let $B_G(v, r) := \{w \in V(G) : d_G(v, w) \leqslant r\}$ denote the radius-r ball in G with center v. The *local density* of a graph G is $\mathrm{ld}(G) := \max\{(|B(v, r)| - 1)/r : r > 0,\, v \in V(G)\}$.[5]

The local density of G provides a lower bound on the bandwidth of G. For any ordering $v_1, \ldots, v_n$ of $V(G)$ with bandwidth b, for each vertex v_i, $B_G(v_i, r) \subseteq \{v_{i-rb}, \ldots, v_{i+rb}\}$, so $|B_G(v_i, r)| \leqslant 2rb+1$ and $\mathrm{ld}(G) \leqslant 2\,\mathrm{bw}(G)$. In 1973, Erdős conjectured that $\mathrm{bw}(G) \leqslant O(\mathrm{ld}(G))$ for every graph G [10, Section 3]. This was disproved by Chvátalová [12] who describes a family of n-vertex trees T with $\mathrm{ld}(T) \leqslant 25/3$ and $\mathrm{bw}(T) \in \Omega(\log n)$.[6] Thus, $\mathrm{bw}(G)$ is not upper bounded by any function of $\mathrm{ld}(G)$, even for trees. This remains true for trees of bounded pathwidth: Chung and Seymour [11] describe a family of n-vertex trees T with local density at most 9, pathwidth 2, and $\mathrm{bw}(T) \in \Omega(\log n/\log\log n)$. On the other hand, in his seminal work, Feige [25] proves that bandwidth is upper bounded by the local density times a polylogarithmic function of the number of vertices.

THEOREM 7. (FEIGE [25]) *For every n-vertex graph G,*

$$\mathrm{bw}(G) \in O\left(\mathrm{ld}(G) \cdot \log^3 n \sqrt{\log n \log\log n}\right) .$$

Rao [33] improves Theorem 7 in the special case of planar graphs:

THEOREM 8. (RAO [33]) *For every n-vertex planar graph G,*

$$\mathrm{bw}(G) \in O\left(\mathrm{ld}(G) \cdot \log^3 n\right) .$$

By Theorem 8, to prove Theorem 1 it suffices to show the following *local sparsification lemma*:

LEMMA 1.3. *Every n-vertex planar graph G has a set X of $O(\sqrt{n}\log^2 n)$ vertices such that*

$$\mathrm{ld}(G - X) \in O(\sqrt{n}/\log n).$$

Lemma 2.1, in Section 2, is a generalization of Lemma 1.3 that trades off the size of X against the local density of $G - X$. The proof of Theorem 1 is concluded by the end of Section 2. The proofs of Theorems 3–6 appear in the full version.

Proving Theorem 6 is the most technically demanding aspect of our work, for reasons that we now explain. Theorem 8 is not stated explicitly in [33]. It is a consequence of the following two results of Feige [25] and Rao [33]. (The definition of (k, η)-volume-preserving contractions is in the full version, but is not needed for the discussion that follows):

THEOREM 9. ([33]) *For every integer $k \in \{2, \ldots, n\}$, every n-vertex planar graph has a $(k, O(\sqrt{\log n}))$-volume-preserving Euclidean contraction.*

[5] The -1 in this definition of local density does not appear in the definitions of local density used in some other works [25, 33], but this makes no difference to our asymptotics results. Our definition makes for cleaner formulas and seems to be more natural. For example, under our definition, the local density of a cycle of length $2k+1$ is 2 and every r-ball contains exactly $2r+1$ vertices for $r \in \{1, \ldots, k\}$. Without the -1, the local density of a cycle is 3, but only because radius-1 balls contain three vertices.

[6] The proof of Theorem 3.4 in [12] constructs an infinite tree with vertex set $\mathbb{N}^2$ that has local density at most 25/3 and infinite bandwidth. In this construction, for each $h \in \mathbb{N}$, the maximal subtree that includes $(0, a_h)$ but not $(0, a_h + 1)$ has $n_h \leqslant 2 \cdot 8^h$ vertices and bandwidth at least $h/9 \in \Omega(\log n_h)$.

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THEOREM 10. ([25]) *For every n-vertex graph G with local density $\mathrm{ld}(G) \leqslant D$ that has a (k,η)-volume-preserving Euclidean contraction,*[7]

$$\mathrm{bw}(G) \in O((nk\log n)^{1/k}\, Dk\eta \log^{3/2} n) \ .$$

Theorem 8 is an immediate consequence of Theorems 9 and 10 with $k = \lceil \log n \rceil$. Unfortunately, we are unable to replace "planar graph" in Theorem 9 with "subgraph of $H \boxtimes P$." The proof of Theorem 9 relies critically on the fact that planar graphs are $K_{3,3}$-minor-free. Specifically, it uses the Klein–Plotkin–Rao (KPR) decomposition [28] of K_h-minor-free graphs G, which partitions $V(G)$ into parts so that the diameter of each part C in G is $\mathrm{diam}_G(C) \in O_h(\Delta)$ (for $O(\log n)$ different values of Δ).[8] Although the KPR decomposition generalizes to K_h-minor-free graphs for fixed h, this does not help because $H \boxtimes P$ is not K_h-minor-free for any fixed h, even when H is a path.

Although $H \boxtimes P$ is not necessarily K_h-minor-free, a very simple (two-step) variant of the KPR decomposition accomplishes some of what we want. That is, it provides a partition of $V(G)$ so that each part C has $\mathrm{diam}_{H\boxtimes P}(C) \in O(\Delta)$. However, distances in G can be much larger than distances in $H \boxtimes P$, so this decomposition does not provide upper bounds on $\mathrm{diam}_G(C)$. To deal with this, we work with distances in $H \boxtimes P$, so that we can use the simple variant of the KPR decomposition.

Working with distances in $H \boxtimes P$ requires that we construct a set X of vertices so that the metric space $\mathcal{M} := (V(G) \setminus X, d_{(H\boxtimes P)-X})$ has local density $O(\sqrt{tn}/\log n)$. That is, we must find a set X of vertices in $H \boxtimes P$ so that radius-r balls in the graph $(H \boxtimes P) - X$ contain at most $rD+1$ vertices of $G - X$, for $D = \sqrt{tn}/\log n$. As it happens, the same method used to prove Lemma 1.3 (the local sparsification lemma for planar graphs) provides such a set X.

However, we are still not done. The simple variant of the KPR decomposition guarantees bounds on $\mathrm{diam}_{H\boxtimes P}(C)$, but does not guarantee bounds on $\mathrm{diam}_{(H\boxtimes P)-X}(C)$, which is what we now need. This is especially problematic because $G - X$ may contain pairs of vertices v and w where $d_{(H\boxtimes P)-X}(v,w)$ is unnecessarily much larger than $d_{H\boxtimes P}(v,w)$. This happens, for example, when vertices added to X to eliminate overly-dense radius-r balls happen to increase the distance between v and w even though no overly-dense radius-r ball contains v and w.

To resolve this problem, we introduce a distance function d^* that mixes distances measured in $H \boxtimes P$ with distance increases intentionally caused by "obstacles" in X. This contracts the shortest path metric on $(H\boxtimes P)-X$ just enough so that, for each part C in (a refinement of) the simplified KPR decomposition, $\mathrm{diam}_{d^*}(C) \in O(\Delta)$. The trick is to do this in such a way that d^* does not contract the metric too much, so the local density of the metric space $\mathcal{M}^* := (V(G) \setminus X, d^*)$ is $O(\sqrt{tn}/\log n)$, just like the metric space $\mathcal{M}$ that it contracts. At this point, we can follow the steps used in Rao's proof to show that the metric space $\mathcal{M}^*$ has a $(k, O(\sqrt{\log n}))$-volume-preserving Euclidean contraction (the equivalent of Theorem 9) and then apply a generalization of Theorem 10 to establish that $G - X$ has bandwidth $O(\sqrt{tn}\log^2 n)$.

2 Local Sparsification

In this section, we prove a generalization of the following result:

LEMMA 2.1. *For any $D \in \mathbb{R}$ with $1 \leqslant D \leqslant n$, every n-vertex planar graph G has a set X of $O((n\log n)/D)$ vertices such that $G - X$ has local density at most D.*

Before continuing, we show how this lemma establishes Theorem 1. First note that Theorem 8 and Lemma 2.1 imply:

COROLLARY 1. *For any $D \in \mathbb{R}$ and $n \in \mathbb{N}$ with $1 \leqslant D \leqslant n$, every n-vertex planar graph G has a set X of $O((n\log n)/D)$ vertices such that $G - X$ has bandwidth at most $O(D\log^3 n)$.*

[7] The precise tradeoff between all these parameters is not stated explicitly in [25], but can be uncovered from Feige's proof, which considers the case where $k = \log n$ and $\eta = \sqrt{\log n}\sqrt{\log n + k\log k}$.

[8] The diameter of a subset $S \subseteq V(G)$ in G is $\mathrm{diam}_G(S) := \max\{d_G(v,w) : v,w \in S\}$. In recent work on coarse graph theory (e.g. [6, 24]), $\mathrm{diam}_G(S)$ is called the 'weak diameter' of S, to distinguish it from the diameter of $G[S]$.

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Theorem 1 follows from Lemma 1.1 and Corollary 1 by taking $D := \sqrt{n}/\log n$.

The proof of Lemma 2.1 makes use of the following fairly standard vertex-weighted separator lemma. Similar results with similar proofs appear in Robertson and Seymour [35], but we provide a proof for the sake of completeness.

LEMMA 2.2. *Let H be a graph; let $\mathcal{T} := (B_x : x \in V(T))$ be a tree decomposition of H; and let $\xi : V(H) \to \mathbb{R}$ be a function that is non-negative on $V(H)$. For any subgraph X of H, let $\xi(X) := \sum_{v \in V(X)} \xi(v)$. Then, for any $c \in \mathbb{N} \setminus \{0\}$, there exists $S \subseteq V(T)$ of size $|S| \leqslant c-1$ such that, for each component X of $H - (\bigcup_{x \in S} B_x)$, $\xi(X) \leqslant \xi(H)/c$.*

Proof. The proof is by induction c. The base case $c = 1$ is trivial, since $S := \emptyset$ satisfies the requirements of the lemma. Now assume $c \geqslant 2$. Root T at some arbitrary vertex r and for each $x \in V(T)$, let T_x denote the subtree of T induced by x and all its descendants. Let $H_x := H[\bigcup_{y \in V(T_x)} B_y]$. Say that a node x of T is *heavy* if $\xi(H_x) \geqslant \xi(H)/c$. Since $c \geqslant 1$, r is heavy, so T contains at least one heavy vertex. Let y be a heavy vertex of T with the property that no child of y is also heavy. Then $H' := H - V(H_y)$ has weight $\xi(H') = \xi(H) - \xi(H_y) \leqslant (1 - 1/c) \cdot \xi(H)$. On the other hand, every component C of $H - V(H') - B_y$ has weight $\xi(C) \leqslant \xi(H)/c$. Apply induction on the graph H' with tree decomposition $\mathcal{T}' := (B_x \cap V(H') : x \in V(T))$ and $c' := c - 1$ to obtain a set S' of size at most $c-2$ such that each component X of $H' - (\bigcup_{x \in S'} B_x)$, has weight $\xi(X) \leqslant \frac{1}{c-1} \cdot (1 - \frac{1}{c}) \cdot \xi(H) = \frac{1}{c} \cdot \xi(H)$. The set $S := S' \cup \{y\}$ satisfies the requirements of the lemma. □

A *layering* $\{L_s : s \in \mathbb{Z}\}$ of a graph G is a collection of pairwise disjoint sets indexed by the integers whose union is $V(G)$ and such that, for each edge vw of G, $v \in L_i$ and $w \in L_j$ implies that $|i - j| \leqslant 1$. For example, if r is a vertex in a connected graph G, and $L_i := \{v \in V(G) : d_G(v, r) = i\}$ for each integer $i \geqslant \mathbb{N}$, then $\{L_i : i \in \mathbb{N}\}$ is a layering of G, called a *BFS layering*. A layering $\{L_s : s \in \mathbb{Z}\}$ is *t-Baker* if, for every $s \in \mathbb{Z}$ and $r \in \mathbb{N}$, $G[L_s \cup \cdots \cup L_{s+r-1}]$ has treewidth at most $rt - 1$. A graph G is *t-Baker* if G has a t-Baker layering. Clearly, if every connected component of G is t-Baker, then G is t-Baker.

Every planar graph is 3-Baker, and for a connected planar graph G, any BFS layering of G is 3-Baker [34]. (This is the property used in Baker's seminal work on approximation algorithms for planar graphs [4].) Thus, Lemma 2.1 is an immediate consequence of the following more general result:

LEMMA 2.3. *For any $D \in \mathbb{R}$ and $n \in \mathbb{N}$ with $1 \leqslant D \leqslant n$, any n-vertex t-Baker graph G contains a set X of at most $(18tn \log n)/D$ vertices such that $G - X$ has local density at most D.*

Proof. Let $\mathcal{L} := \{L_s : s \in \mathbb{Z}\}$ be a t-Baker layering of G. Without loss of generality, assume that $L_i = \emptyset$ for each $i < 0$ and each $i \geqslant n$. For each nonnegative integer i and each integer j, let $G_{i,j} := G[\bigcup_{s=j2^i}^{(j+1)2^i-1} L_s]$, and let $G^+_{i,j} = G[V(G_{i,j-1}) \cup V(G_{i,j}) \cup V(G_{i,j+1})]$. Observe that, for every i, the graphs in $\{G_{i,j}\}_{j \in \mathbb{N}}$ are pairwise vertex disjoint. By the definition of $G^+_{i,j}$, this implies that the graphs in $\{G^+_{i,j}\}_{j \in \mathbb{N}}$ have a total of at most $3n$ vertices.

For each $i \in \{0, \ldots, \lfloor \log n \rfloor - 1\}$ and each j, $G^+_{i,j}$ has treewidth at most $3t \cdot 2^i - 1$, since $\mathcal{L}$ is t-Baker. By Lemma 2.2, with weight function $\xi(v) := 1$ for every $v \in V(G^+_{i,j})$ and $c := \lceil |V(G^+_{i,j})|/(D2^{i-1}) \rceil$, there exists a set $X_{i,j} \subseteq V(G^+_{i,j})$ of size

$$|X_{i,j}| \leqslant 3t \cdot 2^i \cdot (c-1) = 3t \cdot 2^i \cdot \left(\left\lceil \frac{|V(G^+_{i,j})|}{D2^{i-1}} \right\rceil - 1 \right) \leqslant \frac{3t \cdot 2^i \cdot |V(G^+_{i,j})|}{D2^{i-1}} = \frac{6t|V(G^+_{i,j})|}{D}$$

such that each component of $G^+_{i,j} - X_{i,j}$ has at most $|V(G^+_{i,j})|/c \leqslant D2^{i-1}$ vertices. Let

$$X := \bigcup_{i=0}^{\lfloor \log n \rfloor - 1} \bigcup_j X_{i,j}.$$

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Then

$$|X| \leqslant \sum_{i=0}^{\lfloor \log n \rfloor - 1} \sum_j |X_{i,j}| \leqslant \sum_{i=0}^{\lfloor \log n \rfloor - 1} \sum_j \frac{6t|V(G^+_{i,j})|}{D} \leqslant \sum_{i=0}^{\lfloor \log n \rfloor - 1} \frac{18tn}{D} \leqslant \frac{18tn \log n}{D} .$$

Now, consider some ball $B_{G-X}(v,r)$ in $G - X$, let $i = \lceil \log r \rceil$, and let j be the unique integer such that $v \in V(G_{i,j})$. Then $B_{G-X}(v,r)$ is contained in a single component of $G^+_{i,j} - X_{i,j}$, and this component has at most $D2^{i-1} = D2^{\lceil \log r \rceil - 1} \leqslant Dr$ vertices. □

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